

9.3 The Original Class - Graphics

First we will focus on the original set of Graphical components, and then move on to the modern Graphics2D class that comes with Swing. As we said, the original Java 2D API works on the basis of the AWT, Abstract Window Toolkit, which relies on the underlying hardware.

9.3.1 Graphics basics

The Java coordinate system begins in the upper-left-hand side of the screen. The 0,0 location is actually behind the title bar. You use the method `getInsets()` of class `Container` to detect where you can start drawing. There are four measurements that tell you how many pixels away from the edge is the available drawing area.

9.3.2 Graphics context

A graphics object remembers a collection of settings for drawing images and text, such as the font you set or the current color. This collection of settings is known as a Graphics Context. All drawing in Java must go through this graphics object, using the established graphics context. Before you can draw on the screen, you have to acquire it. Naturally, a graphics object takes care of acquiring the graphics context, and uses it to control how information is drawn.

A graphics object encapsulates state variables for:

- the component object on which to draw
- information about “clipping” an object, or
- masking pieces of it.
- current color
- current font
- information about the pixels

A Graphics object has methods for:

- font manipulation,
- drawing
- color manipulation.
- Color

This class defines methods and constants used to manipulate colors. Colors are created from:

- red,
- green, and
- blue components.